GG-S-618D
June 30, 1978
SUPERSEDING
Fed. Spec. GG-S-618C
October 31, 1960

FEDERAL SPECIFICATION

SPHYGMOMANOMETER, ANEROID AND MERCURIAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers aneroid and mercurial sphygmomanometers used for the indirect measurement of arterial blood pressure.

1.2 Classification.

1.2.1 Types, styles, classes, and sizes. Sphygmomanometers shall be furnished in the following types, styles, classes, and sizes as specified (see 6.1):

Manometer Gages - Types and Styles

Type I - Aneroid.

- Style 1 - Pocket.
- Style 2 - Hand.
- Style 3 - Desk.
- Style 4 - Wall Mounted.

Type II - Mercurial.

- Style 5 - Desk Portable.
  - a. Case Model.
  - b. Free Standing.
- Style 6 - Stand Portable.
- Style 7 - Wall Mounted.

Arm Cuff - Classes

- Class A - Bandage Cuff.
- Class B - Metal Hook and Metal Rib Cuff.
- Class C - Hook and Loop Closure Cuff - Synthetic Fabric.
- Class D - Hook and Loop Closure Cuff - Cotton or Cotton Blend.
- Class E - Hook and Loop Closure Cuff - Conductive Fabric.

Pressure Graduations

- Size a - 260 mm Hg
- Size b - 300 mm Hg

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

- Fed. Test Method Std. No. 601 - Rubber, Sampling and Testing.
- Fed. Std. No. 751 - Stitches, Seams and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American National Standards Institute:

ANSI Z70.1 (UDC 615.47) - American National Standard for Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Fire Protection Association Publication:

NFPA Bulletin No. 56A - Standards for the Use of Inhalation Anesthetics.

(Applications for copies should be addressed to National Fire Protective Agency, 470 Atlantic Avenue, Boston, MA 02210.)

American Society for Testing and Materials (ASTM) Standard:

C 162 - Standard Definitions of Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

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3.1.1 Metal parts. Metal parts, other than the carrying case, shall be either corrosion resistant, or rendered corrosion resistant by suitable plating or surface treatment (organic finishing, chemical, or electrochemical). Threads of threaded metal parts shall be treated chemically or electrochemically to render their surfaces corrosion resistant. Finish shall be in accordance with the manufacturer's commercial practice.

3.1.1.1 Pressure control valve. The pressure control valve shall be fabricated of stainless steel, nickel plated brass or chromium over nickel plated brass, or anodized aluminum. Springs or needle valves if made of steel shall be suitably electroplated. Elastomeric components may be present as (1) the facing of a spring loaded pressure plate, (2) an internal nipple slit at the closed end, or (3) an internal nipple or closed tube slit at the sidewall.

3.1.2 Rubber.

3.1.2.1 Rubber tubing. Tubing shall be rubber with a 5/32 inch inside diameter. An accelerated aging test shall be performed as specified in 4.3.5. Rubber tubing shall have a tensile strength of 1100 pounds per square inch minimum before aging.

3.1.2.1.1 Conductive rubber tubing. Conductive tubing shall be rubber and processed to render it conductive and meet National Fire Protection Association Publication #56A for conductive accessories. Conductive tubing shall have tensile strength of 1100 pounds per square inch minimum. After accelerated aging (as outlined in 4.3.5), tensile strength shall not decrease by more than 25 percent. Each length of tubing shall be plainly marked with suitable legend indicating that it is conductive. When furnished as continuous tubing for replacement use, the tubing shall be repeatedly marked, approximately every 6 inches.

3.1.2.2 Inflation bulb and rubber compression bladder. Inflation bulb and compression bladder shall be fabricated of rubber. The bulb and compression bladder shall be fabricated by the dipped latex process or any other process resulting in a seamless product defined as on lacking any overlay or cut within 1/2-inch of any edge.

3.1.2.2.1 Conductive inflation bulb and conductive rubber compression bladder. Conductive inflation bulb and conductive compression bladder shall be fabricated of rubber, suitably compounded, processed or treated to render them conductive and meet National Fire Protection Agency Publication #56A for conductive accessories. Both shall be marked with suitable legend indicating that they are conductive. Tensile strength of the rubber in the conductive compression bladder shall be 3000 pounds per square inch minimum when tested according to 4.3.4. Tensile strength requirements of tubing integral to the conductive bladder (dipped latex tubing) shall not apply.

3.1.2.3 Conductive rubber feet. The conductive rubber used for feet or glides on sphygmomanometer cases or housing shall meet the requirements of National Fire Protection Association Publication #56A. The rubber feet shall be electrically conductive to the extent that the resistance of objects, which are interconnected with the feet, shall not exceed 1 megohm when measured as specified in 4.3.9.

3.1.3 Glass. Glass parts shall be fabricated of clear flint glass or borosilicate glass, and comply with the requirements hereinafter specified. There shall be no ring strain present in the fabricated parts, and any logitudinal strain present shall be faint and highly diffused. Glass shall show no evidence of scumming, chipping, or cracking. Glass cap or crystal covering the dial face shall be shatterproof. (See ASTM C162 for definitions of terms relating to glass and glass products.)

3.1.4 Plastic.

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3.1.4.1 Plastics used in instrument. Plastics shall be suitable for the purpose intended, and shall not crack, flake, peel, or otherwise disintegrate in normal usage. The plastics used in the instrument shall be free of material defects such as chips, cracks or other irregularities which would render it unfit for use.

3.1.5 Fabric.

3.1.5.1 Sateen. Sateen shall be nondistensible and consist of lustrous cotton in satin weave. The sateen shall weigh approximately 3.2 to 3.8 ounces per square yard and have a minimum breaking strength of 150 x 90 pounds.

3.1.5.2 Cotton twill. Cotton twill shall be nondistensible and conform to the requirements of table I. As an alternate, cotton twill 8.6 ounces per square yard, with a minimum breaking strength of 170 x 80 (pounds), 3 x 1 left hand twill may be used.

3.1.5.3 Cotton oxford. Cotton oxford shall be combed yarn conforming to the requirements of table I.

3.1.5.4 Synthetic fabric. The fabric shall weigh approximately 1.8 to 3.4 ounces per square yard, and be of such length and denier as to produce an evenly blended yarn. The fabric shall be made of a bright continuous thread having a minimum breaking strength of 150 x 90 pounds. A urethane coating is permissible on the unexposed surface.

3.1.5.4.1 Conductive fabric. The fabric shall be nondistensible, and shall weigh approximately 1.8 to 8.6 ounces per square yard. A .002 ± .001 inch neoprene coating is required on both sides to meet conductivity requirement per National Fire Protection Association Publication #56A for conductive accessories. The fabric shall have a minimum breaking strength of 150 x 90 pounds.

3.1.5.5 Polyester/cotton. Polyester/cotton cloth shall be 8.5 ± 0.25 ounces per square yard, 104 x 48 thread count, and have a minimum breaking strength of 150 x 90 pounds.

3.1.6 Sewing thread. Thread used for sewing shall be cotton type I, 3 ply, ticket No. 60 or heavier. Thread used shall be of the same color as the material sewed with the thread. With synthetic fabrics, polyester thread shall be used.

3.1.7 Mercury. Mercury used in the mercurial sphygmomanometers shall contain not more than 10 ppm of foreign matter.

TABLE I - Cloth characteristics

Cloth characteristic	Cotton Oxford	Cotton twill
Weight per square yard (minimum) ounces	5.5	7.9
Thread count, warp (per inch minimum)	196	116
Thread count, filling (per inch minimum)	86	56
Breaking strength, warp (pounds)	180	180
Breaking strength, filling (pounds)	80	120
Weave	Oxford	3/1 Twill to right
Yarn ply, warp	2	2
Yarn ply, filling	1	2

3.2 Component parts. Sphygmomanometers shall consist of the following components: An inflation system, a pressure measuring device, and a means of holding the components. These individual components or replacement parts or complete sphygmomanometers as specified in 3.3 may be ordered by the purchaser (see 6.1).

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3.2.1 Inflation system. Inflation system shall consist of a cuff, compression bladder, inflation bulb, check valve, and a pressure control valve. The interior of the inflation system, shall be free from foreign matter (those materials not present by design) and contamination.

3.2.1.1 Cuff. Cuff for sphygmomanometer shall be of the following classes as specified in the procurement document. Cuffs referred to in the following paragraphs as related to dimensional size refer only to the adult size.

3.2.1.1.1 Class A, bandage cuff. Cuff shall be of material specified in 3.1.5.1 or 3.1.5.5. Cuff shall be two thickness of material not less than 37 inches long and 5-1/2 ± 1/4 inches wide for a length of not less than 18 inches, from which point the material may gradually taper to form a tail of such width as may be easily tucked in for securing the arm band on the arm. The band shall be cross-stitched at a point 9-3/4 to 11-1/4 inches from the wide end of the band, to form a pocket for the rubber compression bladder.

3.2.1.1.1.1 Pocket. The pocket shall be provided with two hemmed openings each 1 ± 1/4 inches in length on one of the lengthwise edges of the pocket. Beginning of each hemmed opening shall be located at 1-3/4 ± 1/2 inches and 3 ± 1/2 inches, respectively from either end of the formed pocket. There shall be a hemmed opening 3 ± 1/2 inches in length, centered in the wide end of the band leading into the pocket. The opening may be finished with a tab of such size and shape to conceal the compression bladder when the tab is tucked into the pocket. A single, hemmed opening 2-1/2 ± 1/8 inches in length located 2-7/16 ± 1/8 inches from the end is permitted.

3.2.1.1.1.2 Stitching and seams. All stitching shall be type 301, of Federal Standard No. 751 with not less than 8 stitches per inch. Seam types shall be in accordance with the requirements of Federal Standard No. 751; type SS-a-1 for seaming, and type EF-a-1 for hemming. Openings into the sleeve (for tubing and insertion of bag) shall be reinforced at their closure points by a double row of stitching.

3.2.1.1.2 Class B metal hook and metal rib cuff. The cuff shall be of two thicknesses of material specified in 3.1.1, 3.1.5.1, 3.1.5.2, 3.1.5.3, 3.1.5.4 and 3.1.5.5. The cuff shall be from 20-1/4 inches to 22-3/16 inches long and from 5-3/8 inches to 5-5/8 inches wide.

3.2.1.1.2.1 Pocket. The cuff shall be provided with a pocket not less than 9-7/8 inches long and not less than 5-3/16 inches wide to house the compression bladder. Either one or two suitable hemmed openings shall be provided on the lengthwise edge of the pocket. When one opening is provided, it shall begin at 1-5/8 inches to 2-1/16 inches and end at 4 inches to 4-13/16 inches from either end of the pocket. When two hemmed openings are provided, each should be 1 ± 1/4 inches in length, and both should be located on the same lengthwise edge of the pocket. In the event that the openings described above will not allow the insertion of the rubber compression bladder, there shall be a hemmed opening, 3 ± 1/4 inches in length, centered in the long edge of the pocket finished with a tab or flap of such size and shape to conceal and confine the compression bladder when the tab is tucked into the pocket.

3.2.1.1.2.2 Closure. The cuff shall be provided with at least 15 adjustments by means of a hook (or hooks) and suitable openings to engage the hook (or hooks). The hook(s) shall be mounted on a rib extending 80% of the width of the cuff and the openings shall be holes in ribs extending the full width of the cuff. The ribs used for engagement of the hook(s) shall be partially exposed and adequately secured to the cuff. As an alternate, when specified by the procuring authority, the hook (or hooks) shall be mounted on a rib (or ribs) measuring a minimum of 3-3/8 inches and a maximum of 5-5/8 inches, and the openings shall be holes, in ribs measuring a minimum of 3-3/8 inches and a maximum of 5-5/8 inches. The surface of the cuff, opposite to the surface used for engagement of the hooks shall have the cotton sateen covering over the full width of the cuff.

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read the intervening marks, and each 20 millimeter each 10 millimeter division longer

3.2.1.1.2.3 Stitching and seams. All stitching shall be type 301, of Federal Standard No. 751 with not less than 8 stitches per inch. All stitching shall be backtacked at start and finish, and all openings shall be reinforced by a double row of stitching. Seam types shall be in accordance with the requirements of Federal Standard No. 751; type SS-a-1 for seaming, and type EF-a-1 for heming. Openings into the sleeve (for the tubing and insertion of the bladder) shall be reinforced at their closure points by a double row of stitching.

3.2.1.1.3 Class C hook and loop closure cuff - synthetic fabric. The cuff shall be made of the material specified in 3.1.5.4.

3.2.1.1.3.1 Closure. The contact closure shall consist of two polyester monofilament strips, one of which incorporates a minimum of 93 hooks per square inch and the other containing a minimum of 778 closed loops per square inch of binding. As an alternate, 400/465 hook & pile attachment may be used for closure. The contact pile shall have a minimum of 24 square inches of loops and shall have a minimum of 16 square inches of hooks.

3.2.1.1.3.2 Cuff design. The cuff shall be 19-3/4 to 24 inches long and 5-1/2 ± 1/4 inches wide. It shall provide a pocket, having dimensions of 11 ± 3/4 inches in length by 5-3/8 ± 1/2 inches in width, which shall be suitable for accommodating the compression bladder. The pocket section shall include an opening 2-1/2 ± 1/2 inches which will permit the tubes to protrude from the pocket. The opening shall be reinforced with double stitching and located 2-1/4 to 3-1/4 inches or 5-3/4 to 6-1/4 inches from the cuff width edge. A manometer tab, if present, shall be 1-3/4 ± 1/8 inches in length by 1-1/8 ± 3/16 inches in width. A suitably located artery indexing label or mark shall be present. Female pile calibration marks are permitted. Should the basic material be coated, the coated side shall be on the inside (unexposed) surface of the cuff.

3.2.1.1.3.3 Stitching and seams. All stitching shall be in conformance with Federal Standard No. 751, type 301, with either 8 to 10 or 10 to 12 stitches per inch. Seams shall be type SS-e-2(a) and SS-3-2(b). All stitching shall be backtacked at start and finish, and all openings shall be reinforced by a double row of stitching.

3.2.1.1.4 Class D, hook and loop closure cuff - cotton or cotton blend. The cuff shall be as specified in 3.2.1.1.3 except that the material shall be in accordance with 3.1.5.2, or 3.1.5.3 or 3.1.5.5.

3.2.1.1.5 Class E, hook and loop closure cuff-conductive. The cuff shall be made of suitable, conductive material as specified in 3.1.5.4.1. The conductive cuff may consist of Class A, Class B, Class C or Class D configurations, providing the material is treated in accordance with 3.1.5.4.1. The conductive cuff must be used only with a conductive inflation system: conductive rubber compression bladder 3.1.2.2.1, conductive inflation bulb, 3.1.2.2.1, conductive tubing, 3.1.2.1.1, conductive rubber feet, 3.1.2.3, and other conductive components as applicable.

3.2.1.1.5.1 Closure. Same as provided in Class A, Class B, Class C or Class D configurations.

3.2.1.1.5.2 Cuff design. The conductive cuff shall be constructed as specified in Class A, Class B, Class C, or Class D configurations except that it be marked with suitable legend indicating that it is conductive.

3.2.1.1.5.3 Stitches and seams. Same as provided in Class A, Class B , Class C or Class D configurations.

3.2.1.2 Compression bladder, classes A, B, C, and D. The compression bladder for classes A, B, C, and D shall be made of rubber, as specified in 3.1.2.2 and for class E of rubber as specified in 3.1.2.2.1.

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3.2.1.2.1 Type I, styles 1, 3, and 4 and type II. The compression bladder for use with class A, class B, class C, class D, and class E cuffs of these types, and styles sphygmomanometers shall be 8-5/8 to 10-7/16 inches long and 4-5/8 to 5-1/4 inches wide. The bladder shall be capable of withstanding an internal pressure equal to the maximum gage pressure or without leaking when tested as specified in 4.3.8. The compression bladder when inflated shall not extend beyond the opening in the cuff.

3.2.1.2.1.1 Tubing. Two lengths of latex tubing shall be attached by the dipped latex process, resulting in a seamless product, to one long side of the bag and shall provide air entry to and exit from the bladder. One tube shall be not less than 12 inches nor more than 19 inches long and shall be centered 1-3/4 inches to 2-9/10 inches from one corner of the bladder. The other tube shall be not less than 5 inches long and shall be centered 1-1/4 inches ± 1/4 inch from the first tube. Tube measurements include shoulder portion of the compression bladder. Otherwise, both tubes may be the same length, but not less than 8 inches long. In the event that one tube is shorter than the other, the one tube shall be fitted with a female metal Luer fitting in conformance with American National Standards Institute publication ANSI Z70.1-1, (UDC 615.47), to make a tight fitting connection with the male metal Luer fitting of the extension tube (see 3.2.2.2.3). In the event that both tubes are of equal length, either may be so fitted. This fitting described above is required only for the mercurial sphygmomanometer but may be furnished with the aneroid sphygmomanometer at the manufacturer's option if the design of his equipment so permits. The use of mated metal Luer fittings to increase tube length is permissible.

3.2.1.2.1.2 Physical requirements of compression bladder. The bladder shall conform to the following physical requirements as shown in table II and 3.2.1.2.2.1. The conductive compression bladder shall conform to the requirements of 3.1.2.2.1.

3.2.1.2.2 Type I, style 2. The compression bladder for use with class A, class B, class C, class D, and class E cuffs of the hand model sphygmomanometer shall have exactly the same specifications as specified in 3.2.1.2.1 except that only one length of latex tubing equipped with a female metal Luer lock fitting, in conformance with American National Standards Institute publication ANSI Z70.1 (UDC 615.47), is required. This length of tubing shall be attached by the dipped latex process and shall be centered 1-7/8 inches to 2-3/4 inches from one corner of the bag and shall be not less than 20 inches long. The bag shall conform to the following physical requirements as shown in table II, and 3.2.1.2.2.1.

TABLE II. Physical properties

Tensile stress at 100 percent elongation (maximum) pounds per square inch	Tensile strength (minimum pounds per square inch)		Ultimate elongation (minimum) percent	
	Before aging	After aging	Before aging	After aging
100	4000	3000 See 3.2.1.2.2.1	800	800

3.2.1.2.2.1 Tensile strength after aging. Tensile strength of samples after being subjected to the accelerated aging test shall not decrease by more than 25 percent. The bladder shall be subjected to the accelerated aging test specified in section 4 and upon being so subjected, the bladder shall conform to the above table II and, in addition, shall show no stiffening, softening, cracking, or other changes in the conditionof the bladder. Discoloration of the bladder resulting from this test may be disregarded.

3.2.1.3 Inflation bulb. The inflation bulb shall be fabricated of rubber as specified in 3.1.2.2, except the conductive bulb shall be according to 3.1.2.2.1. Inflation bulb shall be ovoid in shape, approximately 3-1/4 inches long by 1-9/16 inches minimum diameter, 2 inches maximum diameter and capable of making a tight fit with the proximal end of the pressure control valve. One end shall be friction fitted with a metal inlet valve fitted with a stainless steel ball for a check valve, or otherwise fitted with a suitable air intake (check) valve.

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3.2.1.3.1 Physical requirements. Inflating bulb, together with the check valve in place, after being subjected to the accelerated aging test outlined in section 4.3.5 shall show no stiffening, softening, cracking, or other change in the condition of the bulb. Discoloration of the bulb resulting from this test shall be disregarded.

3.2.1.4 Pressure control valve. The pressure control valve shall be approximately 1-1/2 inches in overall length and shall be made of the materials specified in 3.1.1 and 3.1.1.1. The knurled thumb control knob may be made of plastic, providing its threaded inner section which engages the pressure control valve body is made of metal. The pressure control valve shall be capable of holding a maximum gage pressure, or 300 mm whichever is greater, 150 mm, and 50 mm of mercury with a leakage rate not exceeding 12 mm of mercury per minute (the pressure control valve shall be tested at these points and at any intermediate points between 50 mm and 300 mm or maximum gage pressure as may be deemed necessary to determine compliance).

3.2.1.4.1 Type I, styles 1, 3 and 4 and type II valves. The pressure control valve shall consist of an air intake valve located at the proximal end to accommodate the inflation bulb, an air release valve and a nipple located at the distal end to which a tube of the compression bladder may be fitted. See figure 1. The pressure control valve shall allow the flow of pressure impulses from the inflation bulb through the air intake valve and into the compression bladder, but prevent the flow of air in the opposite direction (with the air-release valve in the closed position). The air release valve shall contain a captive knurled and threaded control to permit adjusting the rate of pressure release from the inflated compression bladder. The pressure control valve may be furnished with a suitable replaceable air filter mounted in the incoming air passage at the proximal (bulb) end.

3.2.1.4.1.1 Nipples. The nipple at the proximal end shall have a minimum overall length of 11/32 inch and shall be so constructed that the largest diameter of the nipple shall be 3/8 inch $\pm$ 1/64 inch to provide a tight fit with the inflation bulb. This nipple shall be sufficiently reduced in diameter beyond the point of largest diameter (toward air-release valve) to prevent slipping of the bulb from the nipple after the two are joined. The nipple at the distal end of the pressure control valve shall have a minimum overall length of 13/32 inch and shall be so constructed that the largest diameter of the nipple shall be 9/32 inch $\pm$ 1/32 inch and the nipple shall be sufficiently reduced in diameter beyond the point of largest diameter (toward air-release valve) to prevent slipping of tubing from this nipple after the two are joined.

3.2.1.4.2 Type I, style 2. The pressure control valve shall consist of an air-release valve supplied with a knurled thumb control device, a nipple at the proximal end of the release valve to accommodate the bulb and a metal Luer locking feature consisting of a metal collar with a circular internal thread which shall engage with the female metal Luer fitting (see 3.2.1.2.2) and permit it to be drawn into a securely locked position, threaded into the case at the distal end of the pressure control valve to which the tube of the compression bladder may be connected. The Luer lock shall prevent the gage from becoming disconnected when left suspended from the tube of the compression bladder while the cuff is being applied or removed.

3.2.2 Pressure measuring device.

3.2.2.1 Type I - aneroid.

3.2.2.1.1 Manometer gage, type I, styles 1 and 2 (pocket and hand models). The manometer gage shall be of the aneroid type.

3.2.2.1.1.1 Dial face. The dial face of the gage shall be of metal or plastic, and shall be not less than 1-3/4 inches in diameter. The dial face shall have a pressure scale clearly and permanently marked thereon, the scale extending from zero to not less than 300 millimeters of mercury.

3.2.2.1.1.2 Graduations. The graduations shall cover the entire range from 20 millimeters Hg to not less than 300 millimeters Hg in 2 millimeter divisions, with each 10 millimeter division longer than the intervening marks, and each 20 millimeter division throughout the scale being marked with Arabic numerals indicating the pressure value. There shall be a zero-indicating mark or marks within the limits of which the pointer shall rest when there is no pressure upon the diaphragm or bellows. The zero-indicating mark shall be situated on the dial and the instrument designed such that the dial may be set in only one specifically indicated fixed position.

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3.2.2.1.1.3 Pointer. The pointer of the gage shall be as light as is practicable and of sufficient fineness at the point to allow accurate reading of the pressure The pointer shall be counterbalanced and securely attached to the spindle.

3.2.2.1.1.4 Gage housing. The gage housing shall be made of the material specified in 3.1.1 and of dustproof construction. The housing for type I, style 1 gage shall have a suitable device permanently attached to hold the gage on the cuff when the instrument is in use. The case of the manometer shall be provided with a screw or a bezel retained by a locknut of snap ring and so constructed as to permit the use of a standard watch glass of unbreakable type of shatterproof glass or clear plastic.

3.2.2.1.1.5 Inner mechanism. No zero stop pin shall be used, and there shall be no internal resistance of the pointer movement within 25 angular degrees of the zero-indicating mark. The metal bellows shall be preseasoned for a minimum of 30 days in dead stock to assure stability of the instrument calibration when tested as specified in 4.3.1.1. No screw holding any of the inner mechanism shall be accessible from the outside of the instrument and all parts of the inner mechanism including the connecting tubing between the metal bellows and the external hose nipple shall be fabricated from metal.

3.2.2.1.2 Manometer gage type I, style 3 (desk model). The manometer gage shall be of the aneroid type.

3.2.2.1.2.1 Dial face. The dial face of the gage shall be of metal or plastic, and shall be not less than 3 inches in diameter.

3.2.2.1.2.2 Graduations. The scale shall be graduated from 0 to not less than 300 millimeters of mercury. The graduation shall cover the entire range from 20 millimeters to 300 millimeters/Hg in 2 millimeter divisions with each 10 millimeter division emphasized, and numbered at each 20 millimeter division, throughout the scale. There shall be a zero-indicating mark, in the limits of which the pointers shall rest when there is no pressure upon the diaphragm or bellows. The 0 indicating mark shall not be adjustable and if situated on the dial, the dial shall be nonrotatable.

3.2.2.1.2.3 Pointers. The pointers shall be counterbalanced and securely attached to the spindle.

3.2.2.1.2.4 Inner mechanism. No zero stop pin shall be used, and there shall be no internal resistance of the pointer movement within 25 angular degrees of the zero-indicating mark.

3.2.2.1.3 Manometer gage, type I, style 4 (wall model). The manometer gage shall be of the aneroid type.

3.2.2.1.3.1 Dial face. The dial face of the wall manometer gage shall be of metal or plastic and shall be approximately 6 inches in diameter the dial shall be contoured to bring the graduations near the level of the pointer so as to minimize the parallax phenomenon. The dial face shall have a pressure scale to not less than 300 mm Hg showing a graduation every 2 mm. For ease of reading, the graduations shall be arrayed radially, not less than 2-1/4 inches from the center of the dial. The numerals shall start at 20 mm and appear every 20 mm on the dial. The dial face (and the pointer) shall be covered by a crystal of shatterproof glass or plastic. The dial shall show, the word "Sphygmomanometer," as well as "mm Hg."

3.2.2.1.3.2 Pointer. The pointer shall be counterbalanced and securely attached to the spindle.

3.2.2.1.3.3 Inner mechanism. No zero stop pin shall be used, and there shall be no internal resistance of the pointer movement within 25 angular degrees of the zero indicating mark. This will permit ready determination of the gage accuracy at the zero pressure reading.

3.2.2.2 Type II - mercurial. The manometer shall consist of a calibrated glass or clear nonyellowing plastic manometer tube, a reservoir, a connecting tube or channel for joining the reservoir to the manometer tube, a rubber extension tube if required, and sufficient mercury for the proper operation of the instrument (see 3.1.7). If the reservoir is integral to the unit, a connecting tube is not required.

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3.2.2.2.1 Calibrated manometer tube. The manometer tube shall be constructed of glass or plastic and shall be of adequate length for the purpose intended. The tube shall have an inside diameter of not less than 3.9 mm and shall have a wall thickness of not less than 1.8 mm. The tube shall be linear (straight) and shall be designed to remain so during normal use. The plastic tube shall withstand an aging test as shown in 4.3.6.2. Where required, see 6.1 to specify material preference.

3.2.2.2.1.2 Design. The manometer tube shall be calibrated and so designed in conjunction with the design of the remainder of the instrument as to be interchangeable, without calibration, on instruments of the same design and pressure range of same manufacture. The manometer tube shall be calibrated at an angle identical to its angle of application.

3.2.2.2.1.3 Graduations. The graduations shall be on or immediately adjacent to the manometer tube. If the graduations are not on the manometer tube itself, then graduations must be arranged so as to be visible from both sides of the mercury column. The graduations when on the glass manometer tube shall be engraved and filled with pigment, etched and filled with pigment or accomplished by silk screening and fusing in or fusing on, a pigment. The graduations when on the plastic tube shall be made permanent, either by protection with a protective sleeve or by silk screen printing with an abrasion resistant ink. The pigment shall be red or black in color. The graduations shall extend from zero to not less than 260 millimeters of mercury in 2 millimeter divisions with proportionately longer graduation markings at each 10 millimeter division of the tube. The numerals shall be alternately placed on right and left side of the channel at each 10 millimeter division of the tube. The color with which the numerals are marked shall be in clear contrast to the background of the number place in order to be easily read.

3.2.2.2.1.4 Tube mounting and fittings. The tubes shall be so mounted in the apparatus as to provide a leakproof channel for the mercury and to be easily removable and replaceable for cleaning or for replacement. The upper end of the tube shall be fitted with a removable filter which shall prevent the loss of mercury while still allowing the tube at all times to be open to the atmospheric pressure and which shall also serve to prevent rapid oscillation and fluctuation of the mercury column (i.e., damp the column). The instrument shall be so designed as to prevent accidental disassembly of the tube mounting and fittings so as to eliminate mercury spillage. The filter shall be easily removable and replaceable to allow the addition of mercury, when necessary.

3.2.2.2.2 Reservoir and connecting tube. The reservoir and connecting tube or channel shall be fabricated of metal or plastic providing the material prevents amalgamation with the mercury. The design and dimensions of the reservoirs of individual manufacturers shall be such as to allow interchangeability of the manometer tubes of like manufacturer and the same pressure range (see 3.2.2.2.1). The reservoir shall be fitted with a safety trap to prevent the loss of mercury regardless of the position of the instrument. Where required the top of the reservoir shall be provided with a fitting of proper dimensions to accommodate the rubber extension tube and to make a leak-proof connection therewith. The connecting tube or channel shall provide a mercury-tight connection between the reservoir and the manometer tube.

3.2.2.2.3 Extension tube. When required a rubber extension tube (see 3.1.2.1) not less than 25 inches long shall be provided at the end of the mercury reservoir with a male metal Luer fitting by means of which an air-tight fit may be made into the female metal Luer fitting in conformance with American National Standards Institute publication ANSI Z 70.1 (UDC 615.47), on the free tube of the sphygmomanometer bag.

3.2.2.2.4 Extendable tubing. Extendable (coiled) tubing shall be approximately 4 to 8 feet, depending on the application, when extended. The extendable tubing shall be provided with a male metal Luer fitting, in conformance with American National Standards Institute publication ANSI Z 70.1 (UDC 615.47), by means of which an airtight fit may be made into the female metal Luer fitting, in conformance with American National Standards Institute publication ANSI Z70.1 (UDC 615.47), on the free tube of the compression bladder.

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3.2.3 Case.

3.2.3.1 Type I - Aneroid.

3.2.3.1.1 Styles 1 and 2. The carrying case shall be fabricated of embossed black cowhide weighing 2-1/2 to 3 ounces per square foot or man-made vinyl, suitable for the purpose intended. The case shall have a closure and shall have maximum dimensions of 8 by 4-1/2 by 2-1/2 inches. The case shall hold the complete sphygmomanometer and all of the necessary accessories without bulging or injury to the parts. Sphygmomanometer shall be placed in carrying case with inflation bulb (fitted with pressure control valve) attached.

3.2.3.1.2 Style 3. Desk model shall be housed as specified in 3.3.1.3.

3.2.3.1.3 Style 4. Wall model shall be housed as specified in 3.3.1.4.

3.2.3.2 Type II. Mercurial.

3.2.3.2.1 Styles 5a and 5b. For style 5a sphygmomanometer, desk, case model, the case shall be fabricated of rust-resistant metal suitably coated inside with enamel, and outside with a durable baked enamel finish, of the manufacturer's standard color. The maximum outside dimensions of the case shall not exceed 2 1/4 by 4-1/2 by 13-3/4 inches. The lid of the case shall be hinged on the rear short side and shall be equipped with a device for setting and holding it at right angles with the lower portions of the case. This device shall also prevent the accidental dropping of the lid when the instrument is in use. The case shall be fitted with a press-to-release type catch lock which shall securely keep the case closed. Both the device for holding the lid open and the catch lock for holding the case closed shall be of a type easily replaceable by the user in case of breakage. The complete manometer shall be mounted on the inside cover of the case. Style 5b sphygmomanometer, desk, free-standing, shall be made of material which will withstand accidental impact The overall dimensions shall be 2 to 3 inches by 1-1/4 to 2-1/2 inches by 13 to 16 inches in height, excluding the base plate. This model shall be securely mounted to a suitable base plate.

3.2.3.2.2 Style 6. Style 6 sphygmomanometer, the complete manometer shall be constructed of metal, or molded polystyrene. The manometer shall be securely mounted on a rigid stand. The base may have four rubber feet or free rolling swivel casters for portability. Where required see 6.1 to specify preference. The manometer scale shall be permanently fixed in a vertical or tilted positions. The overall dimensions shall be approximately 38 inches high by 11-1/2 to 22 inches wide at the base. The height may be adjustable by design.

3.2.3.2.3 Style 7. Style 7 sphygmomanometer shall be furnished complete for wall mounting. The manometer shall be mounted on a suitably designed bracket. If required, the manometer tube and reservoir shall be connected to each other at the lower end of the panel by means of the connecting tube. The connections shall be a airtight and mercury tight. The instrument shall provide a place to store the cuff when not in use.

3.2.3.2.4 Styles 5a, 5b, 6 and 7 shall provide an internal reservoir sealed to contain the mercury through normal shipping and use conditions without leaking.

3.3 Assembly.

3.3.1 Type I - Aneroid. Type I aneroid sphygmomanometer shall be furnished in styles 1, 2, 3 and 4, classes A, B, C, D, and E as specified in the procurement document (see 1.2.1).

3.3.1.1 Style 1. Type I, aneroid sphygmomanometer, pocket style, shall be a pocket style instrument complete with an aneroid manometer gage having a dial diameter not less than 1-3/4 inches, suitable class A, B, C, D, or E cuff as specified in procurement document, corresponding compression bladder and inflation bulb with pressure control valve, contained in a pliable leather, or vinyl, case.

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3.3.1.2 Style 2. Type I aneroid sphygmomanometer hand model shall have the manometer gage, inflation bulb, and a pressure control valve incorporated into a single unit which is held in the hand. The unit shall meet all specifications for type I, aneroid sphygmomanometers, pocket style, as covered in 3.2.1.2.2; 3.2.1.4.2; 3.3.1.1. The complete unit shall be supplied with a suitable class A, B, C, D, or E cuff, all of which shall be contained in a pliable lleather or vinyl case.

3.3.1.3 Style 3. Type I aneroid sphygmomanometer, desk model, shall be a desk model instrument having a dial diameter not less than 3-1/4 inches and with an easel base permitting positioning the dial to reading angle best suited to the user. To be supplied complete with aneroid manometer, suitable cuff as specified in procurement document, rubber compression bladder, and one additional tube not less than four feet long or when two tubes are supplied one tube not less than 4 feet long equipped with a male metal Luer fitting on one end, conforming with American National Standards Institute publication ANSI Z70.1 (UDC-615-47), which connects with the female metal Luer fitting on the tube of the compression bladder and the other end fitted with a twist fit metal adapter to engage with the manometer fitting, and an inflating bulb with pressure control valve.

3.3.1.4 Style 4. Type I, aneroid sphygmomanometer, wall model, shall be an instrument housed in a shatterproof plastic case, equipped with a suitable bracket for wall mounting and a place to store the cuff when the instrument is not in use. Wall sphygmomanometer is to be supplied with suitable cuff, corresponding compression bladder, tubing, and inflation bulb with pressure control valve. Wall manometer shall be equipped with a coiled extension tube not less than 4 feet long and equipped with a metal Luer fitting in conformance with ANSI Z70.1 (U.D.C. - 615.47) on one end which connects with female metal Luer fitting on tube of compression bladder.

3.3.2 Type II - mercurial. The mercurial sphygmomanometers shall be furnished in styles 5a, 5b, 6, and 7, classes A, B, C, D, and E and sizes a and b, as specified in the procurement document (see 1.2.1). Each style shall be complete with a mercurial manometer, corresponding cuff as specified in the procurement document, compression bladder, an inflating bulb with pressure control valve, and the necessary accessories for the particular instrument.

3.3.2.1 Style 5a. Style 5a sphygmomanometer shall be supplied in a case fabricated of corrosion resistant metal.

3.3.2.2 Style 6. Style 6, mercurial sphygmomanometer shall be contained in a metal or molded polystyrene body resting on four rubber feet or casters. Style 6 sphygmomanometer may also be polystyrene where the scale shall be permanently fixed in a vertical or tilted position. Conductive rubber feet shall be provided where required.

3.3.2.3 Style 7. Style 7, mercurial sphygmomanometer shall be regular manometer unit, mounted on wood panel or a properly designed swivel bracket to fit the manufacturer's design of the sphygmomanometer. The device shall be such as to allow interchangeability of same manufacturers (see 3.2.2.2.1) of mercurial manometer tubes in instrument of same pressure scale range without recalibration (i.e., instruments of the same range shall be designed to permit the use of interchangeable manometer tubes).

3.3.2.4 Accessories. When specified, see 6.1, a cleaning device shall be supplied with each sphygmomanometer. The device shall be suitable for cleaning the tube. In addition, when specified by the purchaser, the following accessories shall be furnished; extra mercury (see 3.1.7) and extra manometer tubes.

3.4 Accuracy.

3.4.1 Type I - Aneroid. When tested as described in section 4.3.1.1, the error at any point, in any of the tests shall not exceed 3 millimeters Hg. After being subject to pressure or vacuum, change in position of the gage shall not cause the pointer to leave zero limits. The motion of the pointer shall be free from backlash and from irregular motion, when the pressure is changed uniformly. After a sudden jar, the pointer shall return to the point previously occupied. Excessively irregular motion of the pointer shall be considered a sufficient cause for rejection. The pointer shall return to within the zero limits at the end of the tests.

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3.4.2 Type II - mercurial. When tested as described in section 4.3.1.2 the error at any point, in any of the tests, shall not exceed 3 millimeters Hg. The column of mercury shall be free of irregular motion when the pressure is changed uniformly. After a sudden jar the column of mercury shall return to the zero point. Excessively irregular motion or separation of the mercury column shall be considered cause for rejection. The column of mercury shall return to the zero point at the end of tests.

3.5 Leakage of assembled sphygmomanometers. The leakage of the completely assembled sphygmomanometers shall not exceed 15 millimeters Hg per minute (see 4.3.3).

3.6 Barometric pressure. The manometer shall accurately indicate the difference between the pressure in the compression bladder of the instrument and atmospheric pressure, without the necessity for any adjustments in the instrument or manipulation of the manometer.

3.7 Instruction sheets. When specified, see 6.1, each sphygmomanometer shall be furnished with clear instructions for use of the instrument.

3.8 Workmanship. Workmanship shall be in accordance with the manufacturer's commercial practice. Sphygmomanometers shall be free from defects which detract from their appearance or which may impair their serviceability.

3.9 Identification marking. Each sphygmomanometer shall be legibly and permanently marked with the name or trademark of the manufacturer and the manufacturer's serial number where applicable. Separately purchased major components such as compression bladders, cuffs, inflation bulbs, pressure control valves, and gages shall be legibly and permanently marked with the name or trademark of the manufacturer. When specified, for military procurement, components shall be marked as indicated in paragraphs 3.9.1.

3.9.1 Rubber replacement parts. When specified, the inflation bulb and the rubber compression bladder, purchased separately as replacement parts, in addition to the requirements of 3.10, shall be permanently and legibly marked with the date of manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling.

4.2.1 For examination. Unless otherwise instructed by the procuring activity, sampling for examination shall be in accordance with Military Standard MIL-STD-105 to determine compliance with all the requirements contained in this specification.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Military Standard MIL-STD-105 (see table III).

4.2.2.1 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the sphygmomanometers to perform tests as required by the applicable subsidiary specifications or by tests specified herein. The sample size for each test shall be as indicated in table III. Not less than two samples shall be taken regardless of lot size.

4.2.2 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Tests.

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4.3.1 Calibration. All calibration shall be accomplished with a mercurial or quartz bourdon manometer that has been properly standardized and certified by a recognized independent testing laboratory. The instrument employed by the testing laboratory shall be traceable to the NBS Standard.

TABLE III. Sampling Criteria for Tests

Component	Characteristic	Req.	Test procedure	Insp. level	AQL
Rubber tubing	Tensile stress, tensile strength, and ultimate elongation.	3.1.2.1	4.3.4	S-1	1.5
	Accelerated aging test	3.1.2.1	4.3.5	S-2	1.0
Compression bladder	Pressure test	3.2.1.2.1	4.3.8	S-3	0.25
	Tensile strength and ultimate elongation.	3.2.1.2.1	4.3.4	S-1	1.5
	Accelerated aging test	3.2.1.2.2.1	4.3.5	S-1	1.5
Inflation bulb	Accelerated aging test	3.2.1.3	4.3.5	S-1	1.5
Aneroid manometer gage	Calibration tests	3.4.1	4.3.1.1	S-3	1.5
	Full-scale test	3.4	4.3.1.1	S-3	1.5
Mercury manometer tube	Examination	3.1.3 3.1.4.1	4.3.6.1	S-2	1.0
	Aging test	3.2.2.2.1	4.3.6.2	S-1	*
Mercury manometer	Calibration test	3.4.2	4.3.1.2	S-3	0.25
	Full-scale test	3.4	4.3.2	S-3	0.25
Cloth cuff	All requirements in table I	3.1.5	4.3.7	S-1	1.5
Sphygmomanometer	Leakage of completely assembled sphygmanometer	3.5	4.3.3	S-3	1.5

*Acceptance number shall be zero for all sample sizes.

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4.3.1.1 Type I, Aneroid sphygmomanometers. Three calibrations shall be made without tapping and with falling pressures only. The pressure shall be made to fall at a rate of approximately two millimeters per second. The first calibration shall be made only after at least 24 hours have elapsed since the last application of pressure to the instrument. The gage shall be in vertical position. The pressure shall be increased to a value 10 mm Hg above the highest point on the sphygmomanometer scale, then decreased at the prescribed rate in accordance with the calibration procedure. The gage shall be in vertical position. Readings shall be taken with decreasing pressure, starting at the highest point of the sphygmomanometer scale and at all other points which are multiples of 60 millimeters. The pointer of the instrument shall be brought to the first highest division and the pressure read on the calibrating manometer and instrument under test, and recorded. The second calibration shall be done not less than 6 hours after the completion of the first, in the same manner as the first, except that the gage shall be inclined backwards at an angle of 45 degrees with the vertical. After completion of the second calibration, the instrument shall be given 30,000 full scale pulsations at not more than 20 pulsations per minute. The instrument shall then be allowed to rest for 24 hours. At the end of that time it shall be calibrated as in the first test. Each instrument shall meet the provisions of 3.4.1 on each of the three calibration checks. The tests are made without tapping and with falling pressure, since in this manner the conditions of use are most nearly represented. The first calibration is a test of the instrument under the conditions where it is used a few times a day with interval of rest. The second calibration takes into account the fact that the gage is often inclined considerably when attached to the patient's arm. The third test is a measure of ability of the instrument tc retain its calibration after use. By allowing the instrument to rest 24 hours for the temporary elastic effects to disappear, and then recalibrating, any permanent change which may have taken place in the diaphragms or bellows can be detected.

4.3.1.2 Type II - Mercurial sphygmomanometer. One calibration shall be made without tapping and with decreasing pressure only. Readings shall be taken in succession at the highest point of the scale and at all other points which are multiples of 60 millimeters. The readings shall be taken with the pressure falling at a rate of approximately two millimeters per second. Separation of the mercury column shall be cause for rejection. The manometer tube shall be calibrated in the same orientation as its intended use.

4.3.2 Full scale test mercurial sphygmomanometers. After 30 full scale deflections of the column of mercury from 0 to maximum deflection, the manometer shall be calibrated as specified in 4.3.1.2 for differences in the indications of the manometer under test and the standard manometer.

4.3.3 Leak test. Completely assembled sphygmomanometers and valves, when the valves are purchased separately, shall be subjected to an internal air pressure of 50, 150 and 300 mm of mercury or maximum gage pressure whichever is greater. The instrument or valve shall be cut off from the air supply, and the pressure observed for a period sufficient to determine the leakage rate.

4.3.4 Tensile strength and elongation for the rubber compression bladder. These requirements shall be determined by the applicable method outlined in Federal Test Method Standard No. 601, Test Method 7111.

4.3.5 Accelerated aging test. Tubing, compression bladder, and inflation bulb rubber shall be subjected to the accelerated aging test outlined in Test Method 7111 of Federal Test Method Standard No. 601.

4.3.6 Mercury tube.

4.3.6.1 Examination. The manometer tube shall be examined for defects which for would render it unsuitable for its intended use. (see 3.1.3 and 3.1.4.1).

4.3.6.2 Aging test. The plastic tube shall be placed on a horizontal surface in a mechanical convection oven at 190° ± 3°F for one hour without undergoing warpage or any other visible damage.

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4.3.7 Cloth. The cloth used in the cuffs shall be tested and certified by the supplier as being in compliance with specification requirements. The tests shall be conducted in accordance with the supplier's standard commercial practice.

4.3.8 Leakage test (compression bladder). The bladder shall be subjected to a pressure equal to the maximum scale pressure of 300 mm Hg, whichever is greater, for one minute. The bladder shall be secured from ballooning by the use of screening or platens of sufficient strength to withstand the pressure.

4.3.9 Electrical conductivity. The electrical conductivity of parts required to be conductive, as specified herein, shall be tested with an ohmmeter which will maintain a voltage between 90 and 500 volts across a resistance of 1 megohm.

4.4 Examination of preparation for delivery. An examination shall be performed to determine compliance with the packaging, packing, unitization, and marking requirements of section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for examination at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quanity	Quanity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components. such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Complete unit Type I. The sphygmomanometer and instruction sheet when required shall be clean and dry and neatly arranged in the carrying case to prevent the tubing from being kinked or permanently deformed. The carrying case shall be closed and secured and placed in a plastic bag and closed by heat sealing. Sphygmomanometers supplied without, a case shall be clean and dry and neatly arranged and wrapped in tissue or other suitable non-corrosible material.

5.1.1.2 Complete unit-Type II. Each sphygomomanometer shall be clean, dry and have the mercury withdrawn from the glass tube and contained within the reservoir. The glass tube shall be sealed to prevent influx of the mercury during shipment and storage. When units are furnished with acrylic tubes, removal of the mercury is not required. Units furnished with a case shall have the bulb, cuff and tubing neatly arranged and wrapped with tissue or other suitable non-corrosive material.

5.1.1.3 Component part. The valve, tube, or other component part, as specified (see 6.1), shall be packaged in a container to provide protection from contamination and physical damage with rigid or resilient dunnage as necessary to prevent free movement. The cuff when procured separately shall be clean, dry, neatly arranged and packaged in a snug fitting heat sealed or folded flap plastic bag.

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5.1.2 Commercial. The sphygmomanometer and instruction sheet or component part, as specified (see 6.1), shall be packaged in accordance with normal commercial practice. The complete package shall be designed to prevent the item against corrosion, deteriosation or physical damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.1).

5.2.1 Level A. The number of complete units or component parts of one description (as type, class, style, and size) specified (see 6.1) shall be packed in a wood box with cushioning and a water resistant liner in accordance with the National Motor Freight Classification or the Uniform Freight Classification.

5.2.2 Commercial. The packaged cuffs or sphygomomanometers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. The packing shall also provide for safe redistribution by the initial receiving agency, and shall be acceptable by common carrier under National Motor Freight classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. The package, shipping containers, and unitized loads (when applicable) shall be marked as specified in the purchase order or contract.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Types, styles, classes, and sizes of complete sphygmomanometers required (see 1.2.1 and 3.3).
(c) Replacement parts required (see 3.2).
(d) Accessories required (see 3.3.2.4).
(e) Whether glass or plastic manometer tube is required (see 3.2.2.2.1).
(f) Whether rubber feet or swivel casters are required (see 3.2.3.2.2.).
(g) Packaging and packing required (see 5.1 and 5.2).
(h) Marking required (see 5.4).

6.2 Purchasers should require a guarantee covering defects and accuracy by inserting the following clauses in the contracts:

6.2.1 Defects. The contractor shall furnish with each instrument a certificate that the instrument is guaranteed against all workmanship and material defects for a period of one year from the date of purchase.

6.2.2 Accuracy. The contractor shall furnish with each instrument a certificate or guarantee that the manometer will remain accurate in normal use, and further, that if it should be thrown out of adjustment it will be corrected by the contractor without additional cost to the Government for a period of 10 years from date of acceptance of delivery.

6.3 This specification does not include all types, classes, grades, sizes, etc., of sphygmomanometer, which are commercially available, but is intended to cover the types, etc., normally procured for Government department requirements.

Preparing Activity:

GSA-FSS

Review Interest:

Army - MD
Navy - MS
Air Force - 03
Defense Medical Review Bd. DOD-MB
Directorate of Medical Matl. DM

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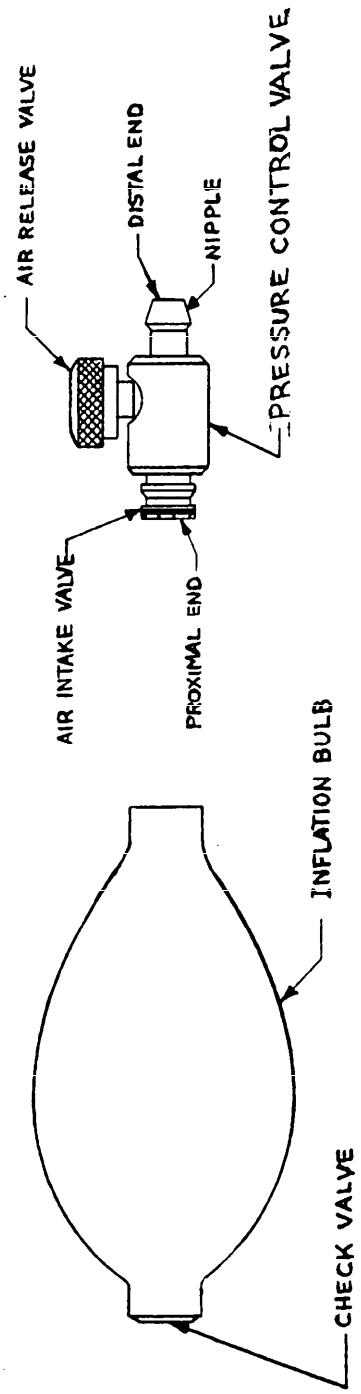


Figure 1 - Inflation Bulb and Pressure Control Valve

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